

# Alessandro Sanvito

SOFTWARE AND AI ENGINEER

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## Experience

### Optiver

Amsterdam, Netherlands

SOFTWARE ENGINEER

August 2023 - Current

- Working in the Data Driven Algorithmic Position Taking team, developing and maintaining data pipelines and trading strategies.
- Building and optimizing data processing systems using Python, Spark, Polars, and DuckDB, with a focus on efficient data storage and retrieval from PostgreSQL and Delta tables.
- Collaborating closely with quantitative researchers and traders to implement and optimize systematic trading strategies, with direct impact on desk PnL.

### Mercedes-Benz (AI-SEE Project)

Stuttgart, Germany

AI RESEARCH INTERN AND MASTER THESIS STUDENT

May 2022 - June 2023

- Conducted cutting-edge research on 3D human avatar modeling from monocular video data in diverse real-world environments.
- Designed and implemented generative neural models, including NeRFs and Diffusion Models, to improve computer vision performance in adverse weather conditions.
- Collaborated with a cross-functional R&D team to refine innovative ideas, resulting in a publication at ICCV (International Conference on Computer Vision).
- Leveraged skills in deep learning, 3D modeling, and generative AI to push the boundaries of digital character creation and visualization.

## Education

### KTH Royal Institute of Technology

Stockholm, Sweden

MSC. IN ICT INNOVATION, DATA SCIENCE

Sept. 2020 - May 2023

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.
- Reproduced data mining-related papers in Python and Java.

### Polytechnic University of Milan

Milan, Italy

MSC. IN COMPUTER SCIENCE AND ENGINEERING

Sept. 2020 - Jul. 2023

- Final grade: 110/110 cum laude
- Developed a winning ML solution on the Twitter dataset for the international RecSys Challenge 2021 with the university team under the supervision of prof. Paolo Cremonesi with Dask, Catboost, and XGBoost.
- Built a recommender system for a book catalogue with NumPy, SciPy, and Pandas.

### Polytechnic University of Milan

Milan, Italy

BSC. IN ENGINEERING OF COMPUTING SYSTEMS

Sept. 2017 - Jul. 2020

- Final grade: 109/110
- Implemented a cardboard game in Java, performing extensive testing with JUnit and Mockito.
- Created a performance oriented graph manipulation tool in C.

## Publications

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 2024 | <b>HINT: Learning Complete Human Neural Representations from Limited Viewpoints</b> , Sanvito Alessandro, Ramazzina Andrea, Walz Stefanie, Bijelic Mario, Heide Felix                                                                                                                                                       | <a href="#">IEEE IV 2024</a>  |
| 2023 | <b>ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings</b> , Ramazzina Andrea, Bijelic Mario, Walz Stefanie, Sanvito Alessandro, Scheuble Dominik, Heide Felix                                                                                                                                  | <a href="#">ICCV 2023</a>     |
| 2022 | <b>United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources</b> , Maldini Pietro, Sanvito Alessandro, Surricchio Mattia                                                                                                                                          | <a href="#">ACM recsys'22</a> |
| 2021 | <b>Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment</b> , Carminati Luca, Lodigiani Giacomo, Maldini Pietro, Meta Samuele, Metaj Stiven, Pisa Arcangelo, Sanvito Alessandro, Surricchio Mattia, Maurera Fernando B. Pérez, Bernardis Cesare, Ferrari Dacrema Maurizio | <a href="#">ACM recsys'21</a> |

## Skills

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|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>              | Python, SQL, Java, C++, C, LaTeX                                                                                                      |
| <b>Data Processing</b>          | Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy                                                                          |
| <b>Machine Learning</b>         | Deep Learning (PyTorch, TensorFlow), Computer Vision, Generative AI (NeRFs, Diffusion Models), scikit-learn, MLlib, Catboost, XGBoost |
| <b>Tools &amp; Technologies</b> | Git, Docker, Linux, CI/CD, pytest                                                                                                     |